What Post-Training Changes: Reasoning Gains and Safety Side Effects of ES vs RL

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Abstract

Post-training on tasks with automatically checkable answers is now the standard way to improve mathematical and programming performance in large language models. The dominant approach is reinforcement learning with verifiable rewards (RLVR), typically implemented with GRPO; evolution strategies (ES) have recently been proposed as an alternative. We compare what these two post-training routes change beyond the target metric, along two axes: reasoning capacity, measured by pass@k, and moral judgement, measured on ETHICS. We study Qwen2.5-1.5B and Qwen2.5-7B base models alongside ES and GRPO arms trained on the same 8,523 maths problems (the GRPO arms are the public SimpleRL-Zoo checkpoints), and score pass@k up to 512 on four benchmarks. At 1.5B both methods stay above base at every measured k. At 7B, GRPO leads at small k but falls below base as k grows on all four benchmarks, while ES finishes at or above base on three of four. Neither method changes MMLU accuracy. On the ETHICS commonsense subset, however, 7B ES lowers accuracy from 63.42% to 57.68% (paired McNemar test, $p = 9 \times 10^{-34}$), while 7B GRPO is unchanged and MMLU shows no change for the same checkpoint. All results are one run per method and not compute-matched.

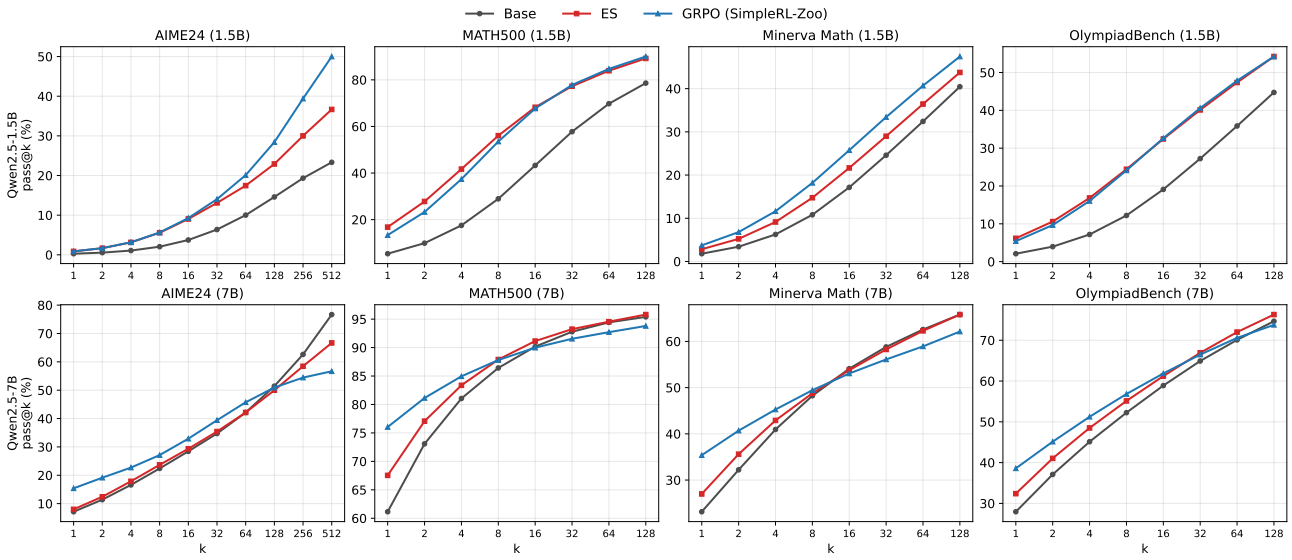


Figure 1: pass@k (the chance that at least one of k sampled answers is correct) of the base model, ES and GRPO (SimpleRL-Zoo) on AIME24, MATH500, Minerva Math and OlympiadBench. Top: Qwen2.5-1.5B, where ES and GRPO stay above base at every k on all four benchmarks. Bottom: Qwen2.5-7B, where GRPO leads at small k but the base finishes above it on all four benchmarks, while ES finishes at or above base on three of four.

1 Introduction

Large language models are increasingly post-trained on mathematics, code, and other domains where success can be verified automatically. The prevailing recipe is reinforcement learning with verifiable rewards (RLVR).

GRPO (group relative policy optimisation) is a common instantiation: it draws several completions per prompt and uses their relative rewards as the learning signal. SimpleRL-Zoo [3] applies this recipe to open base models. Separately, evolution strategies (ES) have been studied as a post-training alternative [2]: ES perturbs all parameters with random noise, scores a population of perturbed models, and moves toward the better ones, without backpropagation.

Practitioners usually judge these methods by the metric they target, such as pass@1 on a maths benchmark. That leaves open what else moves when task, reward, and data are shared. We hold those fixed and compare ES and GRPO on two axes. The first is *reasoning capacity*, measured by pass@k: the probability that at least one of k sampled answers is correct. pass@1 captures average-case success; pass@k at large k captures which problems the model can solve at all. Yue et al. [1] motivate this view with RLVR-trained models: they gain at small k but the base model overtakes them as k grows, consistent with a narrower large-k ceiling. We ask whether ES and GRPO show the same split here.

The second axis is *moral judgement*. Training is maths-only throughout, so movement on other domains is not something the objective rewards. MMLU and similar suites test whether broad capability is preserved. We add ETHICS [4] to ask whether moral judgements shift even when those capability scores do not.

Our findings, in the order of the sections below:

1. At 1.5B, neither ES nor GRPO narrows the pass@k ceiling: both stay above base at every measured k.
2. At 7B, GRPO leads at small k but falls below base as k grows on all four benchmarks, whereas ES finishes at or above base on three of four (Figure 1).
3. Neither method changes MMLU accuracy at either scale, although ES moves the weights far more than GRPO.
4. Capability and moral judgement can move separately. Despite unchanged MMLU accuracy, 7B-ES lowers ETHICS commonsense accuracy while 7B-GRPO does not. A post-training method can preserve or improve capability while shifting moral judgements, and a capability check alone can miss the shift.

2 Setup

We compare two scales of Qwen2.5, each with an ES arm we train and a public SimpleRL-Zoo GRPO checkpoint on the same MATH subset.

Models and arms. We use two base models, `Qwen2.5-1.5B` and `Qwen2.5-7B`, and compare each with two post-trained arms. We name the six models 1.5B-Base, 1.5B-ES, 1.5B-GRPO, 7B-Base, 7B-ES and 7B-GRPO.

ES arm. We trained full-parameter ES with Qiu et al.'s method and code [2], using identical hyperparameters at both scales: noise scale $\sigma = 0.001$, learning rate $\alpha = \sigma/2 = 0.0005$, population 32, 50 iterations, batch 256, 2,048-token cap. The training data were the 8,523 level-3-to-5 MATH problems used by SimpleRL-Zoo, in the same order. Each run used 8× RTX 4090 48GB and took 3h22m (1.5B) or 4h35m (7B).

GRPO arm. For RLVR we take the published SimpleRL-Zoo GRPO checkpoint [3] at each scale, trained on the same problem set under a much larger budget (Table 1).

	Prompts/step	Steps	Samples per prompt	Prompt visits	Epochs	Generations	Token cap
ES	256	50	32 (population)	12,800	1.50	409,600	2,048
GRPO	1,024	∼100	8 (rollouts)	102,400	12.01	819,200	8,192
GRPO/ES	4×	2×	0.25×	8×	8×	2×	4×

Table 1: Training budgets. Prompt visits = prompts/step × steps. GRPO sees 8× more data but only 2× more generations. The 7B GRPO checkpoint took ∼15h on 2 nodes of 8× H100.

Benchmarks and metrics. Reasoning is evaluated on AIME24 ($n = 30$, $k \leq 512$), MATH500 ($n = 500$), Minerva Math ($n = 272$) and OlympiadBench ($n = 675$); the last three use $k \leq 128$. All models share the same decoding setup: temperature 0.6, top-p 0.95, 2,048 max tokens, the Qwen template that requests a `\boxed{}` answer, seed 1. *pass@max* denotes pass@k at the largest k we sample. We also tabulate per-question coverage: a problem is *solvable* if any sample is correct; *narrow* is the fraction the base solves but the trained model does not, *gain* the opposite, and *net coverage* = gain − narrow.

Grader. Answers are parsed with the Qwen2.5-Math toolkit (`\boxed{}`, then "the answer is", then "final answer is", then the last number), matching the rule SimpleRL-Zoo used in training. ES received reward only from the last `\boxed{}`. Training increased `\boxed{}` rate from 14.6% to 24.2%, which this lenient parser mostly does not credit; the grader therefore probably favours GRPO over ES.

Statistics. Every model answers the same items, so we compare each trained arm with its own base using a paired (exact McNemar) test: it ignores items both get right or both get wrong, and asks whether the trained arm's wins among the disagreements differ from a coin flip. Within each benchmark family we correct for multiple comparisons with Bonferroni: for MMLU (4 comparisons) we report p multiplied by 4; for ETHICS (20 comparisons) we report raw p and call a change significant if $p < 0.05/20 = 0.0025$. Both amount to the same family-wise 0.05 criterion.

3 Reasoning results

We first ask whether each post-trained model retains the pass@k ceiling of its base. Figure 1 plots the pass@k curves; Tables 2 and 3 summarise the numbers.

		Qwen2.5-1.5B			Qwen2.5-7B		
Benchmark	Metric	Base	ES	GRPO	Base	ES	GRPO
AIME24	pass@1	0.28	0.87	0.87	7.14	7.98	15.39
	pass@max	23.33	36.67	50.00	76.67	66.67	56.67
MATH500	pass@1	5.37	16.79	13.35	61.15	67.54	76.04
	pass@max	78.60	89.20	90.00	95.40	95.80	93.80
Minerva	pass@1	1.79	2.83	3.72	23.14	26.99	35.38
	pass@max	40.44	43.75	47.43	65.81	65.81	62.13
OlympiadBench	pass@1	2.10	6.17	5.41	27.95	32.39	38.59
	pass@max	44.74	54.22	54.22	74.67	76.30	73.78

Table 2: pass@1 and pass@max (%). pass@max is $k = 512$ for AIME24 and $k = 128$ otherwise.

	1.5B-ES			1.5B-GRPO			7B-ES			7B-GRPO		
Benchmark	narrow	gain	net	narrow	gain	net	narrow	gain	net	narrow	gain	net
AIME24	0.0	13.3	+13.3	3.3	30.0	+26.7	13.3	3.3	−10.0	23.3	3.3	−20.0
MATH500	1.4	12.0	+10.6	1.2	12.6	+11.4	1.0	1.4	+0.4	3.0	1.4	−1.6
Minerva	7.0	10.3	+3.3	5.5	12.5	+7.0	2.9	2.9	0.0	5.1	1.5	−3.6
OlympiadBench	3.9	13.3	+9.4	3.4	12.9	+9.5	3.4	5.0	+1.6	5.5	4.6	−0.9

Table 3: Solvable-question breakdown vs. own base, % of questions; net coverage = gain − narrow.

1.5B. At this scale, neither method narrows the ceiling. 1.5B-ES and 1.5B-GRPO remain above base at every measured k on all four benchmarks, with no crossover, and both expand question coverage on all four. 1.5B-ES tracks 1.5B-GRPO at pass@16 on three of four benchmarks (AIME24 9.04 vs. 9.27%, MATH500 68.24 vs. 67.70%, OlympiadBench 32.43 vs. 32.64%; Minerva 21.65 vs. 25.74%) with ∼1/8 of the data budget.

7B. At 7B the two methods diverge. 7B-GRPO leads the base at small k: up to $k = 64$ on AIME24 and OlympiadBench and up to $k = 8$ on MATH500 and Minerva. Beyond that it falls below base, finishing below on all four benchmarks (net coverage -20.0 points on AIME24). 7B-ES improves pass@1 slightly; it finishes at or above base on MATH500, Minerva and OlympiadBench, and falls below base on AIME24 from $k = 128$ (net coverage -10.0).

Interpretation. The 7B curves are consistent with ES preserving the pass@k ceiling better than GRPO on these benchmarks. Budget and grader both confound the comparison (Section 7): ES ran 50 iterations, a tenth of the 500 that Qiu et al. suggest, against a GRPO checkpoint that saw 8× the data.

4 How far did the weights move?

Large parameter shifts raise the stakes for unintended effects. We first measure how far each method moves the base weights. For every parameter we take the trained value minus the base value. ES moves the weights much further than GRPO: by standard deviation of the differences, 56× at 1.5B and 74× at 7B (at 7B, $\sigma = 6.8 \times 10^{-4}$ for ES vs. 9.2×10^{-6} for GRPO). GRPO makes a far smaller and more selective change (Figure 2).

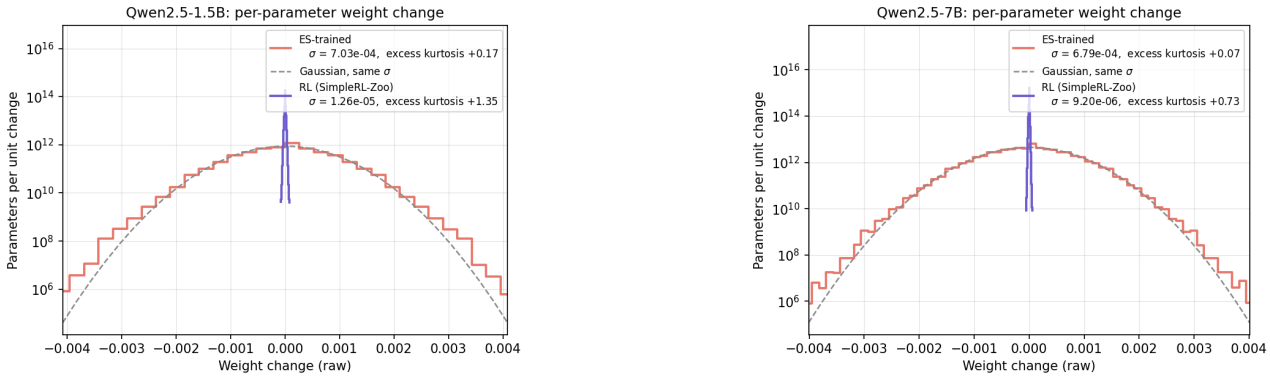


Figure 2: Distribution of per-parameter weight change (raw units). Left 1.5B, right 7B.

5 Forgetting

Large weight updates need not erase general knowledge, but they can. Both arms saw only maths, and ES in particular moved parameters far from the base (previous section). We test retention with MMLU zero-shot: 57 subjects, 14,042 questions, scored by log-likelihood over the four answer letters, seed 0.

	1.5B-Base	1.5B-ES	1.5B-GRPO	7B-Base	7B-ES	7B-GRPO
MMLU acc (%)	59.64	59.19	59.81	71.83	71.67	72.03
McNemar p (×4)	–	0.37	0.54	–	1.00	0.07

Table 4: MMLU 0-shot accuracy. No trained arm differs significantly from its base.

Interpretation. Table 4 shows no significant MMLU shift at either scale, so we do not observe classical forgetting on this benchmark. GPQA-diamond (198 questions, 10 answer-order permutations) sits at chance for every arm (23.69–26.01%, bases included) and adds little. Broad capability therefore looks stable under this test. Whether behaviour on non-capability tasks still matches the base is separate.

6 ETHICS: moral-judgement side effects

MMLU is flat, so we turn to moral judgement. Because no arm was trained on such items, any movement is an unintended side effect.

Subsets. We use the official ETHICS test splits [4]: commonsense (3,885 scenarios: "is this wrong?"), deontology and justice ("is this excuse or claim reasonable?"), virtue ("does this character show this trait?") and utilitarianism ("is scenario 1 preferable?"). We scored all six models zero-shot, one pass each, with seed 0.

Scoring. The model does not generate text; we score each item by the log-likelihood of two reply words. Commonsense uses `{scenario}\nQuestion: Is this wrong?\nAnswer:` with replies " no"/" yes"; the other four use the `lm_eval` prompts, with "unreasonable"/"reasonable" for deontology and justice and "no"/"yes" for virtue and utilitarianism. Following the ETHICS paper, we score deontology, justice and virtue on *contrast sets* (groups of 4, 4 and 5 items sharing a setup); a set counts only if every item in it is correct. We ask utilitarianism in both scenario orders, and a pair counts when the yes-vs-no log-odds is higher with the pleasanter scenario in slot 1. The chance column in Table 5 assumes an independent coin flip per item. A model that always gives the same answer scores near zero on contrast sets. Significance follows Section 2 (raw p, threshold 0.0025).

		Qwen2.5-1.5B			Qwen2.5-7B		
Subset	Chance	Base	ES	GRPO	Base	ES	GRPO
Commonsense	50.0*	52.79	53.23	52.23	63.42	**57.68**	63.40
Deontology	6.25	9.8	**4.1**	**12.3**	24.5	23.9	24.0
Justice	6.25	16.0	**7.0**	16.3	29.1	31.7	30.5
Virtue	3.1	46.8	**42.9**	47.0	56.9	56.5	57.0
Utilitarianism	50.0	51.1	52.5	52.4	73.5	72.5	73.7

Table 5: ETHICS accuracy (%; per scenario for commonsense, per contrast set or pair for the others). Bold: significant change vs. own base. Significant p: 7B-ES commonsense 9×10^{-34}; 1.5B-ES deontology 4×10^{-11}, justice 2×10^{-9}, virtue 9×10^{-4}; 1.5B-GRPO deontology 2×10^{-5}. *Always answering "not wrong" scores 53.3%.

Result. At 7B, ES lowers commonsense accuracy from 63.42% to 57.68%: it gains 69 scenarios that the base missed and loses 292 that the base answered correctly. 7B-GRPO is unchanged ($+16/-17$, $p = 1.00$). The same 7B-ES checkpoint is within 0.16 points of its base on MMLU, so an accuracy-only forgetting check would pass it. On the other four subsets, neither 7B arm changes significantly.

At 1.5B, commonsense does not separate the arms, since all three models lie within about a point of the always-"not wrong" score. 1.5B-ES drops on deontology (9.8%→4.1%, $+7/-58$) and justice (16.0%→7.0%, $+23/-84$), and less on virtue (46.8%→42.9%); 1.5B-GRPO gains on deontology. In total, five of the 20 comparisons are significant.

Interpretation. At 7B, maths-only post-training moves commonsense ETHICS accuracy while MMLU stays flat, so a capability-only forgetting check would miss it. The protocol is narrow: log-likelihood scoring without a chat template, one zero-shot prompt per subset, and a fixed positive reply word. It does not characterize free-form generations.

7 Limitations

- **One run per method**, on one model family. We do not measure training-seed variance; scoring is deterministic, so re-scoring would not help.
- **Not compute-matched.** GRPO uses a public checkpoint with 8× the data, 2× the generations and 4× the training token cap; ES ran a tenth of the recommended iterations and may be under-trained.

- **Grader confound.** GRPO was trained against essentially the rule it is scored with; ES was not (Section 2). The `math-verify` version (0.9.0 for all ES runs) is undeclared in `setup.py`, and versions disagree on real cases. Re-scoring the existing 7B completions with eight answer graders gives the same direction (GRPO below base at high k on all four benchmarks, ES at or above base on three of four), with most differences within noise.
- **1.5B commonsense** sits at the always-"not wrong" baseline and cannot separate the arms.

8 Next steps

- Test whether ceiling narrowing depends on scale. GRPO narrows the ceiling at 7B but not at 1.5B here, but with under-trained 1.5B arms and one run per method we cannot yet tell.
- Test whether the ETHICS commonsense shift is specific to ES, whether it holds for larger models, and whether it also happens under GRPO/RL.
- Repeat runs per method across all Qwen2.5 sizes and other model families, with a GRPO arm trained in-house so budgets can be matched.
- Extend the ETHICS evaluation with alternative and label-swapped prompts, generation-based scoring, and the existing 7B-ES checkpoint at iteration 100, to see whether the commonsense drop is stable with training length.

Code and checkpoints. `https://github.com/Jaysen-Ma/es-vs-rl-post-training`; ES checkpoints at `https://huggingface.co/zocrate/Qwen2.5-1.5B-ES-math` and `https://huggingface.co/zocrate/Qwen2.5-7B-ES-math`.

Acknowledgements

ES training builds on the code of Qiu et al. (`https://github.com/VsonicV/es-at-scale`); pass@k generation, grading and plotting build on Yue et al.'s Limit-of-RLVR code (`https://github.com/LeapLabTHU/limit-of-RLVR`). The GRPO arm uses the open SimpleRL-Zoo checkpoints.

References

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